PULLBACK ALONG A MORPHISM OF SCHEMES PRESERVES CONSTRUCTIBLE SHEAVES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for all three categories of sheaves: both reductions are stated with their named inputs and the required partition is constructed explicitly by pullback.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Constructible sheaf on an etale site. Same chapter, the definition of a constructible sheaf of sets, of abelian groups and of Lambda-modules on Y_etale. Supplies the notion preserved by the pullback.
- Locality of constructibility. Same chapter, Lemma lemma-constructible-local: constructibility of a sheaf may be checked after base change to the members of a covering. Reduces the assertion to the case of affine X and Y.
- Characterization of constructible sheaves by partitions. Same chapter, Lemma lemma-constructible-quasi-compact-quasi-separated: over quasi-compact quasi-separated schemes constructibility is equivalent to being finite locally constant on the pieces of a finite constructible partition. Turns the assertion into the production of a finite partition.
- Pullback preserves locally constant sheaves. Same chapter, Lemma lemma-pullback-locally-constant. Shows that the pulled back partition does the job.

Source reference. Original lemma 095F: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 11789–11789. Navigation pointer invoked at proof

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line 11816; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 095E: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 11717–11717. Navigation pointer invoked at proof line 11780; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 095A: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 9960–9960. Navigation pointer invoked at proof line 11824; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. If $\mathcal{F}$ is a constructible sheaf of sets, abelian groups, or Λ-modules (with Λ Noetherian) on $Y_{étale}$, the same is true for $f^{-1}\mathcal{F}$ on $X_{étale}$.*

Proof. By Lemma 095F this reduces to the case where X and Y are affine. By Lemma 095E it suffices to find a finite partition of X by constructible locally closed subschemes such that $f^{-1}\mathcal{F}$ is finite locally constant on each of them. To find it we just pull back the partition of Y adapted to $\mathcal{F}$ and use Lemma 095A. □